EMBEDDED SYSTEM WITH IOT
EXPERIMENT NO. 1

# IoT based Automated Street Lighting System

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Group No.: INDIVIDUAL

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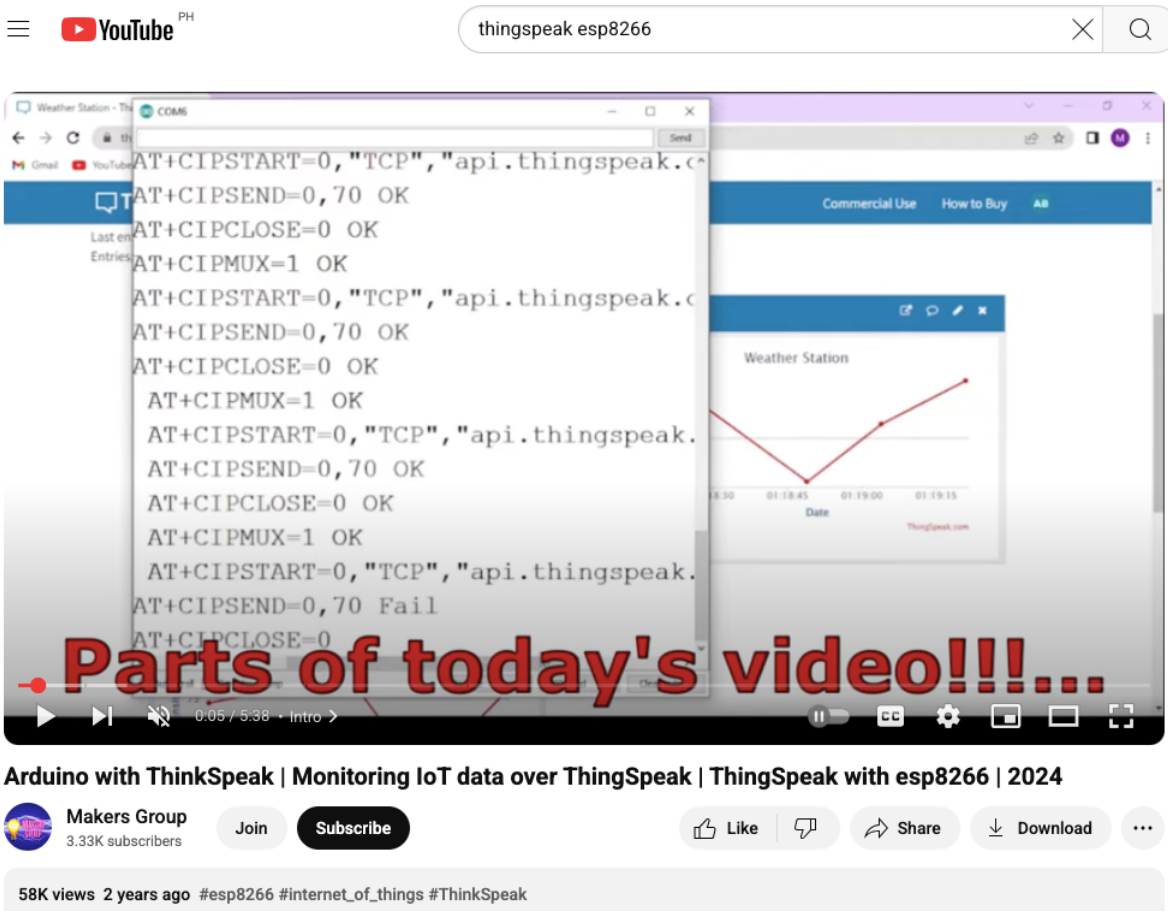
# DISCUSSION

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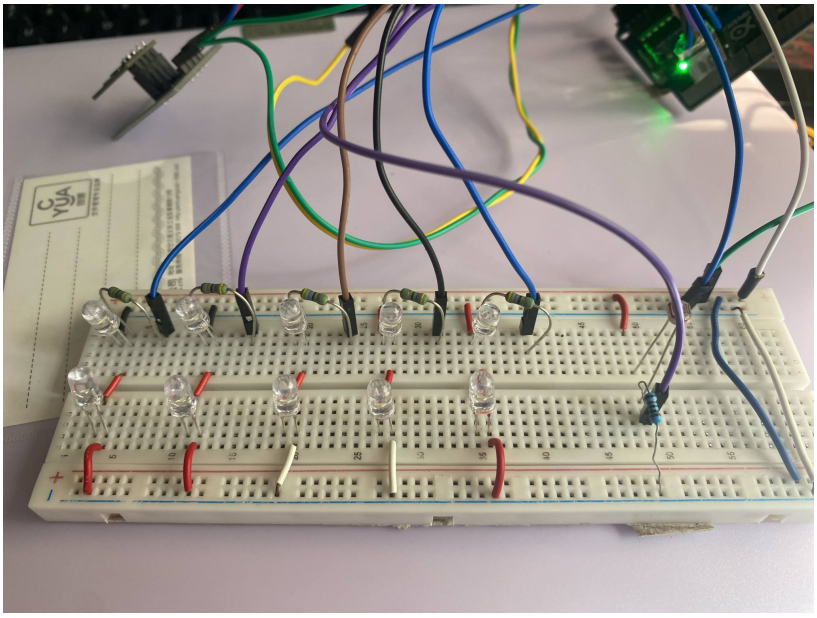
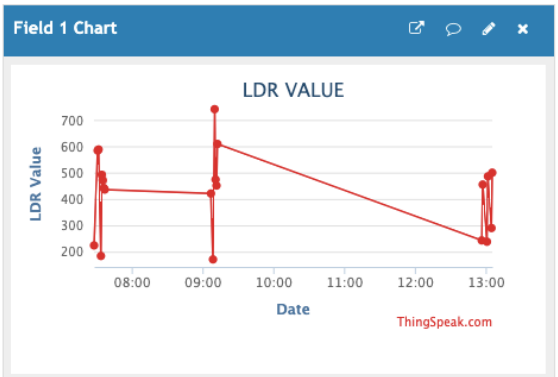
Experiment 1 (IoT Based Automated Street Lighting System) revolves around the introduction of using ThingSpeak with ESP8266 wifi module for Arduino. As I m not familiar with the use of ThingSpeak and the ESP8266 I have to search and watch some guide videos to get me thru the

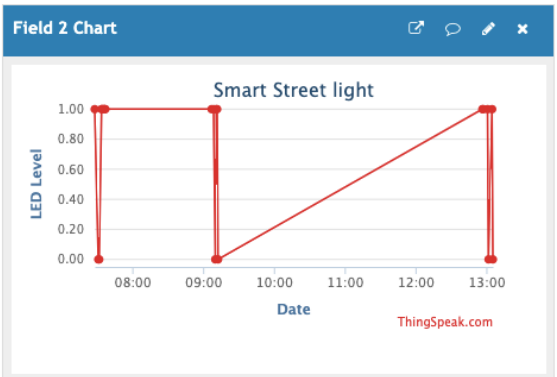
experiment. This allows me to understand some commands that I utilize in this experiment such as commands that allows the wifi module to connect to my specific wifi. Although some part of connecting from wifi module to ThingSpeak is still confusing on my end, i still learned some valuable knowledge that im willing to further enchance.



Once the wifi is integrated, and ready to be work with ThingSpeak. I assembled the leds that I want to utilize to be on and off using the photoresistor, the value of this photoresistor will

be the one to be sent to the ThingSpeak together with the led status if its on or off. The leds will turn on whenever the photoresistor value is less than 400 indicating that no sunlight is detected, while if value is greater than 400 leds will turn off. This logic can be analyze on the ThingSpeak graph.




The data provided in the ThingSpeak graph, is whenever I activate and open the Arduino uno. this is from atleast 7am up to now at around 1:30pm. Field 1 shows the changes of ldr value in different test(20 entries) while the field 2 shows the led state if its in that specific ldr value (also 20

entries). Whenever the Arduino updates the ldr value every 15seconds, the wifi would send this data to ThingSpeak allowing us to check previous test and have sort of like database on testing. The figure shows that we have achieve the objective we wanted, the led will turn on whenever no sunlight is detected, that is around less than 400 ldr value then the led will turn off when sunlight is detected which is greater than 400 ldr value.

# CONCLUSION

The experiment established the idea of an IoT-based Automated Street Lighting System by using an ESP8266 WiFi module and an Arduino to transmit sensor readings to ThingSpeak. With hands-on usage, the system could respond to light levels in the environment as detected by a photoresistor, switching on LEDs automatically when the light value dropped below 400, and switching them off when light was adequate perhaps using a flashlight effectively simulating street lighting control based on sunlight availability.

Although there was initial unfamiliarity with ThingSpeak and ESP8266 configurations, the

process enhanced understanding of wireless communication commands, sensor data handling, and cloud monitoring. The experiment allowed for practical application of concepts from logic circuits, microprocessors, and embedded systems, and added real-time IoT data visualization and logging. Ultimately, the objectives were met since the system performed well and data was received and visualized accordingly on ThingSpeak, offering a foundation for further improvement.